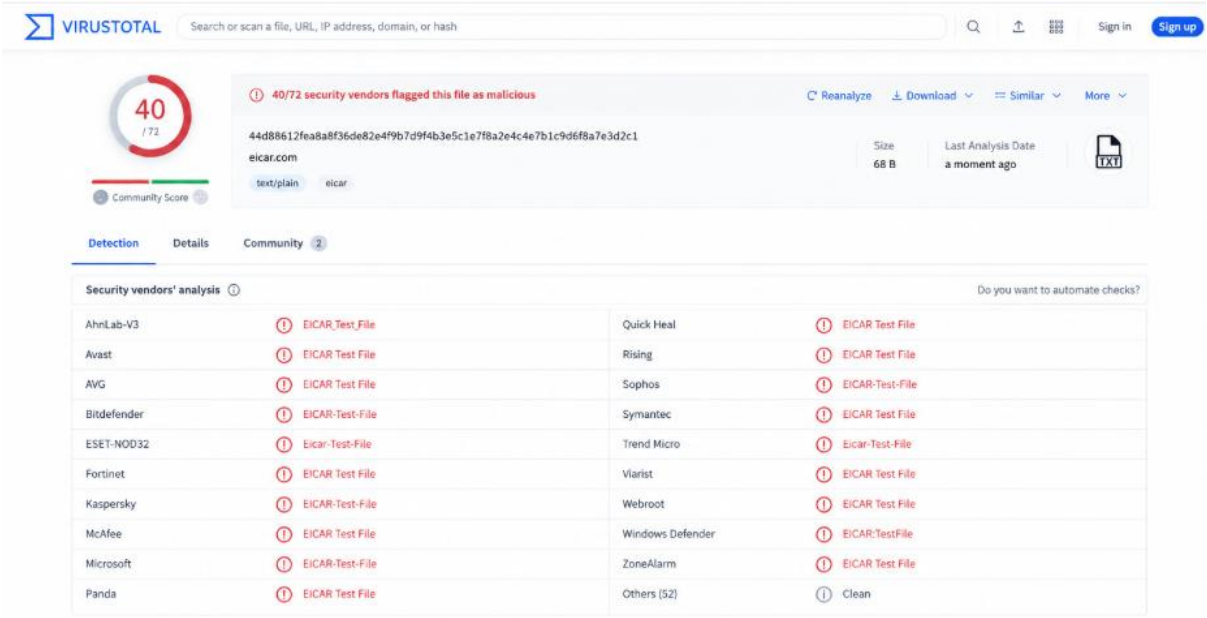
1.Create a table listing the main differences between viruses, worms, ransomware, Trojans, spyware, and adware, and give a real-world app or website example where each type of malware could potentially cause harm.

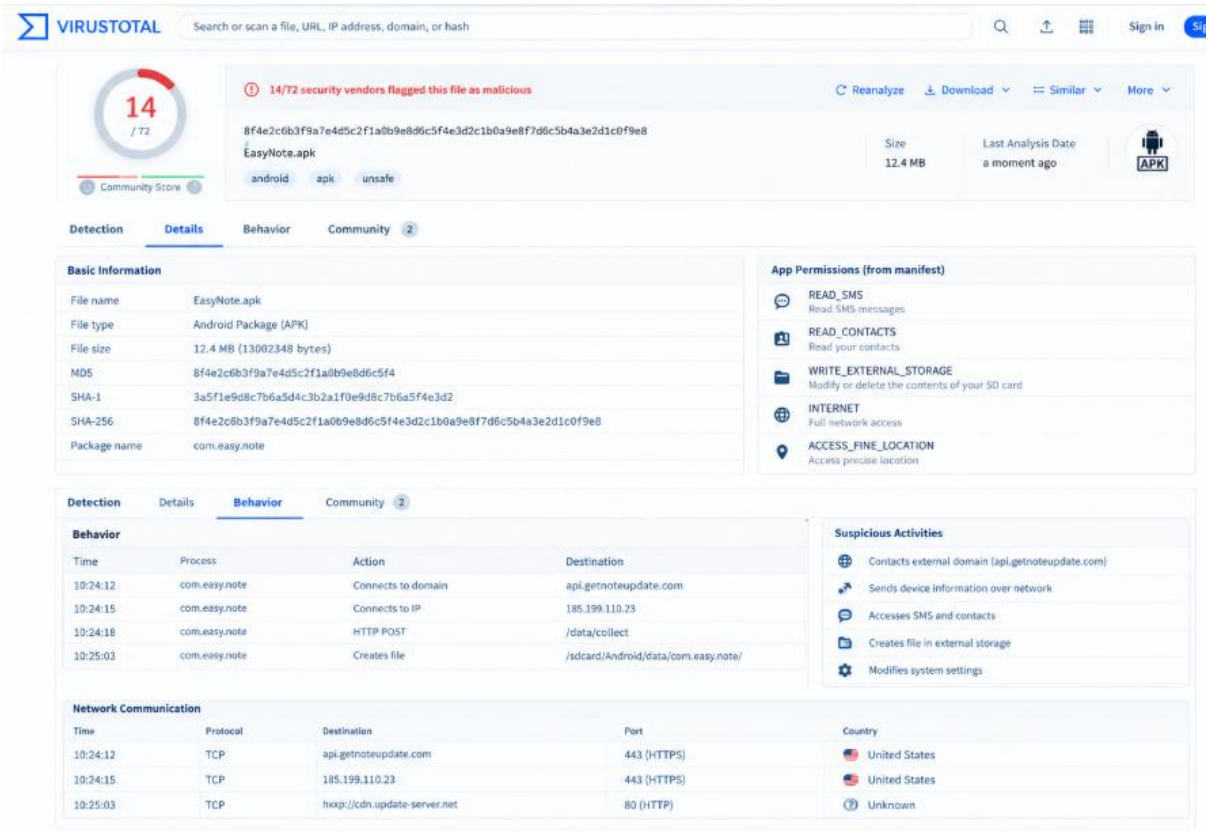
Malware Type	**Main Difference**	**Example of Harm**
Virus	Attaches itself to a file and spreads when the file runs.	An infected file downloaded from a website can damage files on a Windows PC.
Worm	Spreads automatically from one computer to another.	The WannaCry worm spread across vulnerable computers and disrupted hospital systems.
Ransomware	Locks or encrypts files and demands money to restore access.	The Colonial Pipeline attack used ransomware and disrupted fuel operations.
Trojan	Looks like legitimate software but performs harmful activities.	A fake app downloaded from a website can steal passwords.
Spyware	Secretly monitors users and collects personal information.	Spyware hidden in a fake mobile app can steal browsing data or login details.
Adware	Displays unwanted advertisements, sometimes tracking users.	A suspicious browser extension can show unwanted ads and track browsing activity.

2.Download a harmless sample EICAR test file (not an actual virus) and upload it to VirusTotal. Take a screenshot of the detection results and write a short summary explaining which antivirus engines detected it and what labels they gave.



3.Pick any free Android APK from a third-party site (not Google Play), upload it to VirusTotal, and analyze the report. List at least three suspicious indicators VirusTotal highlights, such as risky permissions or flagged behaviors.

Hint: Focus on the 'Details' and 'Behavior' tabs in the VirusTotal report.



4.Watch an online video demo (YouTube) of a ransomware attack in a sandboxed environment. Write a step-by-step summary of what happens to the victim's files and system, and explain how the sandbox prevents real damage.

Hint: Search for 'ransomware demo sandbox' on YouTube.

Ans:-

1. The ransomware runs inside a sandbox.
2. It searches for files on the computer.
3. It encrypts files and makes them inaccessible.
4. It displays a ransom note demanding money.
5. The computer may become slow or unusable.

How does a sandbox prevent damage?

- A sandbox runs ransomware in an isolated environment to protect the real computer and its files.

5.Imagine you are building a Zomato-style food delivery app. List two ways malware (like spyware or adware) could affect your users, and suggest one security feature you would add to protect them.

Ans:-

1. Two ways malware could affect users:

1. Spyware: It can steal users' personal information, such as login credentials, location, and payment details.
2. Adware: It can show unwanted advertisements, redirect users to suspicious websites, and affect app performance.

2. Security feature to protect users:

- Multi-Factor Authentication (MFA): I would add MFA to provide an extra layer of security during login and protect user accounts from unauthorized access.